4_popquiz Power, Wattage

This is a preview of the published version of the quiz

Started: Mar 22 at 7:04pm

Quiz Instructions

Only one attempt. No makeup. Discuss with your classmates

Remember the power of a load of resistance R is P = V^2 /R or P = RI^2 (or you can also use P=V I)

P = V I is always true. The other equations are only true for load of resistance R

Question 1 20 pts

A toaster has a resistance of 75 ohms and is plugged into a 115V outlet.
How much current does it draw ?

Hint : see the equations in the header of the popquiz

- 2 A
- 652 mA
- 1.5 A
- 8625 mA

Question 2 20 pts

A toaster has a resistance of 75 ohms and is plugged into a 115V outlet. Use previous result.

How much does it cost if it operates for a total of 10 hours? About?

The cost of electricity is $0.11 per KWh.

hint: $1 = 100 cents so $0.11 is 11 cents ! Be careful. its tricky

- $2
- 20 cents
- $20
- $1

Question 3 15 pts

A 40W light bulb is plugged in an outlet 120V. Its resistance is?

Hint the equations in the header of the popquiz . Find an equation with V, P and R

- 360ohms
- 4800ohms
- 100 ohms
- 3 ohms

Question 4 20 pts

a 100 W light bulb is plugged into a 120 V outlet. What is its resistance R

- 12000 ohms
- 14,400
- 1.2 ohms
- 144ohms

Question 5 1 pts

Which one has a larger resistance

40W light bulb or 100W light bulb .

See previous questions.

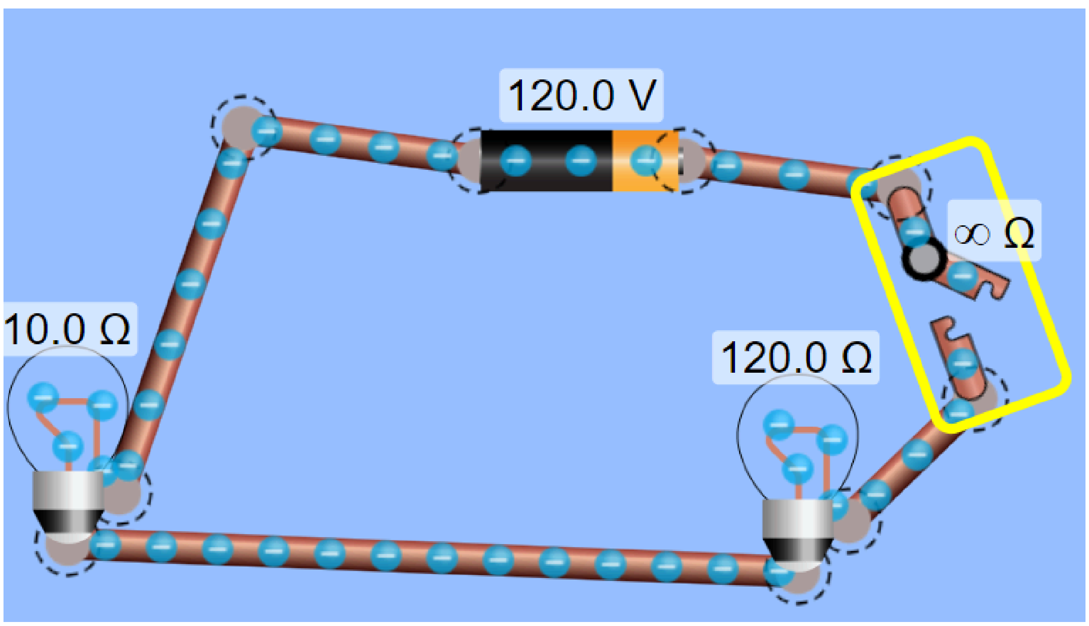
- Can't be determined
- They have the same resistance
- 100W
- 40W

Question 6 15 pts

Go here.

https://phet.colorado.edu/en/simulations/circuit-construction-kit-dc
(https://phet.colorado.edu/en/simulations/circuit-construction-kit-dc)

Have 2 light bulbs in series with a 120V power supply. give one light bulb a very large resistance of 120 ohms and give the other one a small resistance of 10 ohms



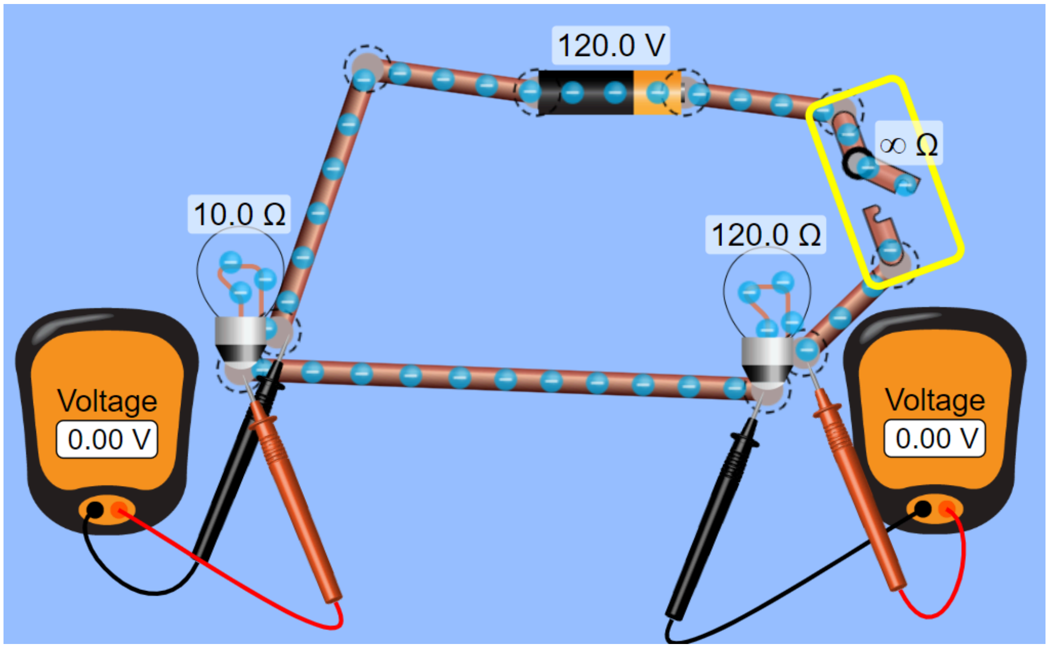
Close the switch. Which one is the brightest ? Isn't that suprising?

- the 10 ohms is the brightest
- The 120 ohms is the brightest
- They are equally bright

IDK

Question 7 5 pts

Use the same circuit and measure the voltage across the loads:



close the switch. What is true

- the voltage across the 10 ohms is way larger
- Both load get 120V across
- they have about the same voltage drop
- The voltage across the 120 ohms is way larger

Question 8 5 pts

based on your investigations above. If you have a 40W light bulb and a 100W light bulb in series. (so 120V in series with those 2 light bulbs) then

- the 40W bulb will be the brigthest
- My brain is foggy
- They have the same bightness
- The 100W will be the brightest

Not saved Submit Quiz